PROBLEM STATEMENT

Federated Learning Platform for Seasonal & Widespread Disease Surveillance

Main Problem: Current disease surveillance systems are **CENTRALIZED, SLOW, & REACTIVE,** leading to **delayed outbreak detection** and **ineffective response.**

KEY ISSUES

DELAYED OUTBREAK DETECTION

Data analyzed daily/weekly → **Late Action**

✕ Missed early warning signs

PRIVACY & SECURITY RISKS

Patient data sharing required → **Data Breaches**

✕ Violates GDPR, HIPAA

LACK OF COLLABORATION

Hospitals work in silos → **No Shared Insights**

✕ Limited detection accuracy

NO REAL-TIME MONITORING

Static reports only → **No Live Tracking**

✕ Cannot detect fast spread

LIMITED AI & PREDICTION

Reactive systems **No Early Prediction**

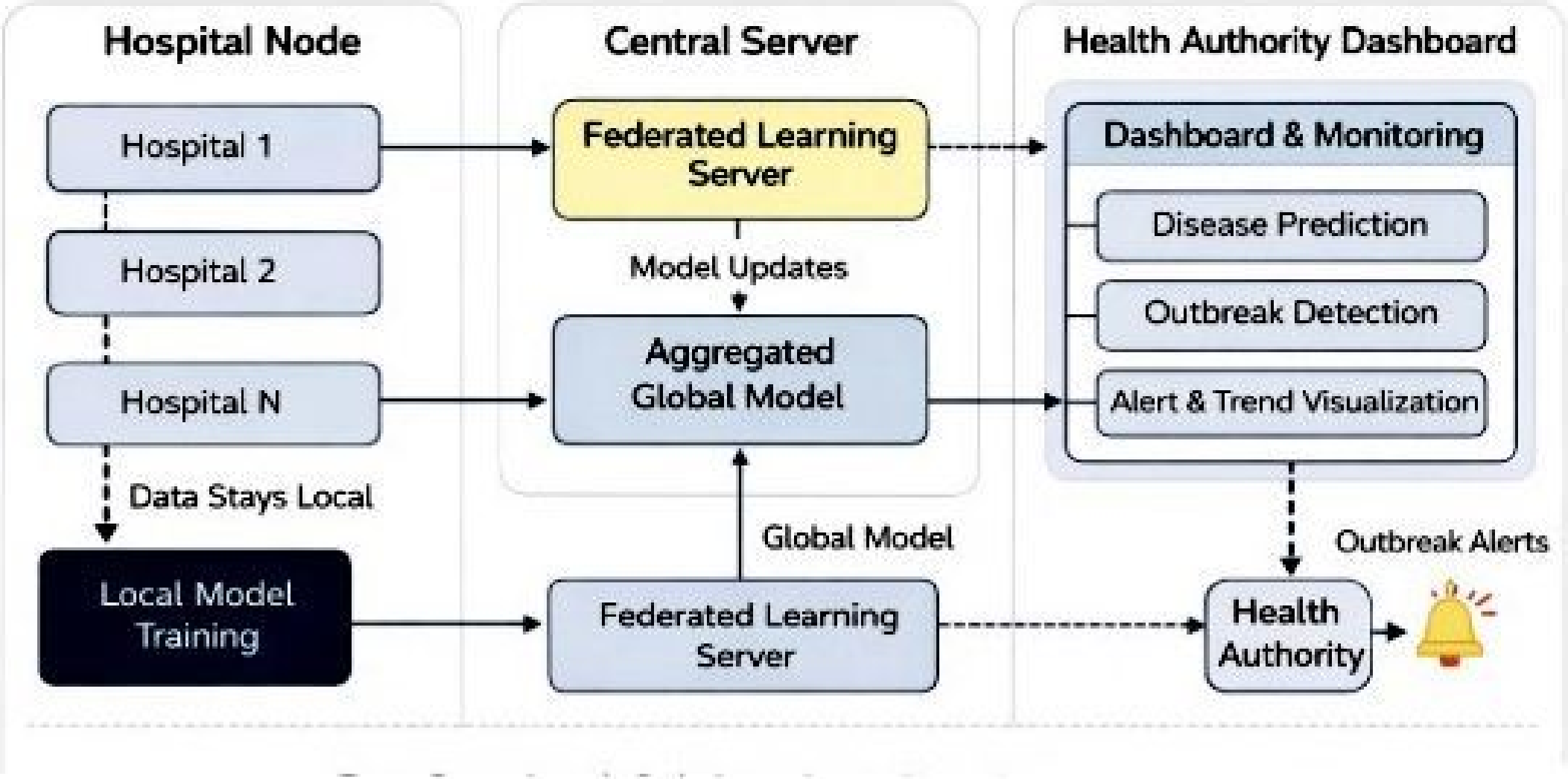
✕ Outbreak detected too late

FINAL PROBLEM STATEMENT

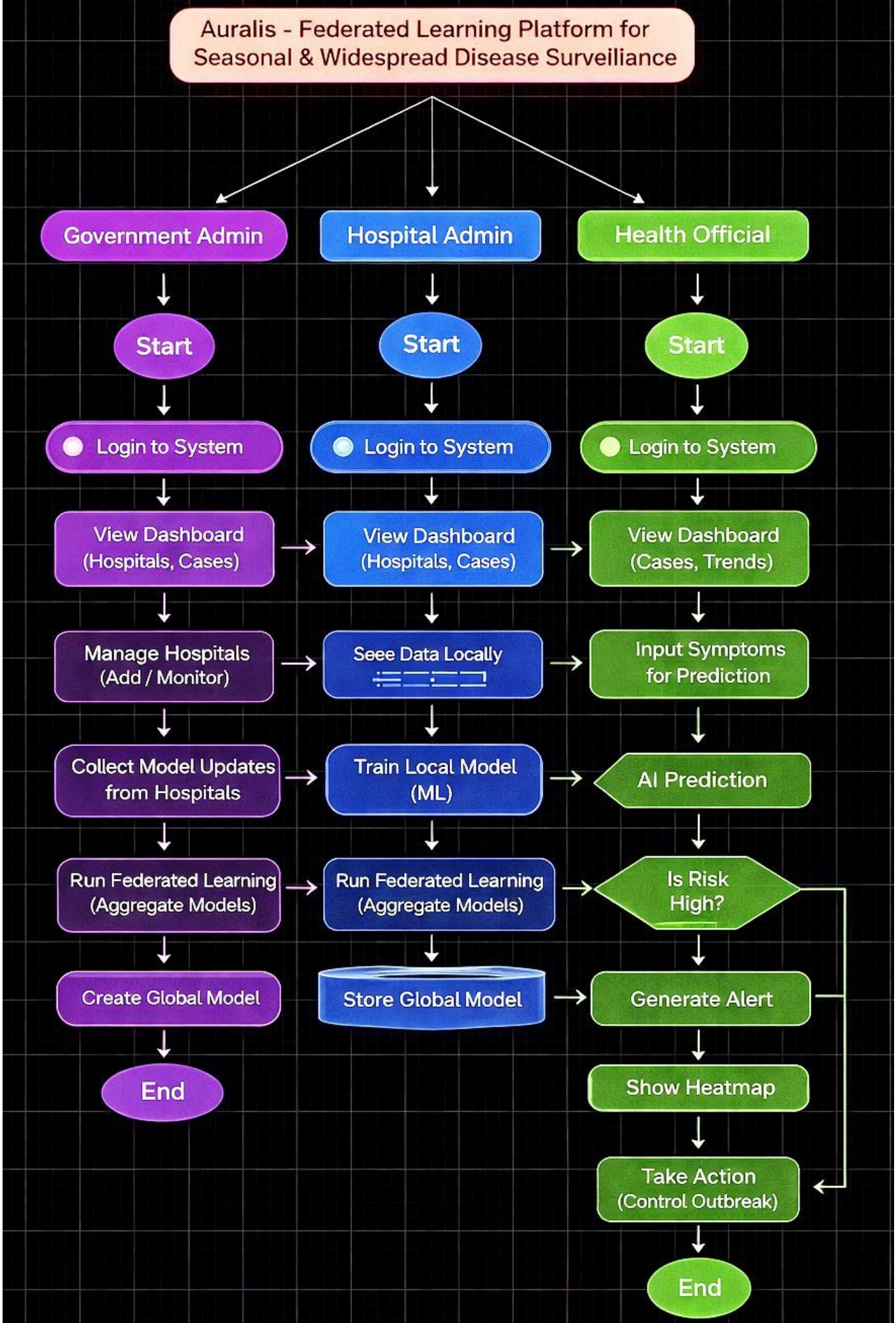
"Existing disease surveillance systems fail to provide **real-time**, **privacy-preserving**, and **predictive** capabilities, resulting in **delayed outbreak detection** and **inefficient healthcare response.**"

Not Predictive | Privacy Invasive | Slow & Reactive | Centralized & Insecure

Architecture Diagram



Work Flow Diagram



Federated Learning Workflow

1 Step 1: Local Training

Hospital 1

Local Data (150 cases)
↓
Train Local Model
↓
Model Weights

Hospital 2

Local Data (150 cases)
↓
Train Local Model
↓
Model Weights

Hospital 3

Local Data (150 cases)
↓
Train Local Model
↓
Model Weights

2 Step 2: Weight Aggregation

Model W1 | Model W2 | Model W3

↓

Federated Averaging
Global W = (W1 + W2 + W3) / 3

↓

Global Model Weights

3 Step 3: Global Model Distribution

Global Model

↓

Hospital 1 (can use global model)

Hospital 2 (can use global model)

Hospital 3 (can use global model)

WHO BENEFITS FROM THIS PROJECT?

Hospitals & Healthcare Providers

- Predict patient inflow in advance
- Prepare: Beds
  - ICU facilities
  - Medical staff
- Improve **treatment planning**

Government & Health Authorities

- Monitor disease spread **in real-time**
- Identify: High-risk zones
  Outbreak regions
- Take early **actions:**
  - Issue alerts
  - Implement containment strategies
  - Plan vaccination drives

General Public / Patients

- Reduced infection risk
- **Saves lives** ❤️

Impact:

- Better public health management
- Faster response to outbreaks

Pharmacies & Medical Suppliers

- Predict demand for medicines
- Stock medicines in advance

Impact:

- Reduced infection risk
- **Saves lives** 💚

Researchers & Data Scientists

- Access improved trained models
- Analyze disease patterns
- Study outbreak trends

Impact:

- Prevents shortages
- Ensures availability during outbreaks

Healthcare System (Overall)

- Better coordination between institutions
- Efficient resource allocation

Impact:

- Stronger and **smarter** healthcare ecosystem

CONCLUSION

This project **benefits hospitals, governments,** the public, and the **entire healthcare ecosystem** by enabling early **disease detection**, better planning, and improved healthcare decision-making.